Julius Zhang

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I am interested in identifying structures and developing new primitives for practical problems.

Experience

Columbia University Researcher, June. 2026 - Present

- Conduct research and attend seminars in the Department of Mathematics.
- Homotopy theory, arithmetic geometry, cryptography, security, and other topics.

a16z Researcher, Apr. 2025 - Present

- Building Jolt, open-source zero-knowledge virtual machine in Rust.
- Implemented and benchmarked torus compression; proved optimality with a conceptual reframing via Hilbert 90.
- Audited proof systems security: verify correctness of elliptic-curve optimizations and formalize folklore results into rigorous proofs.
- Evaluated lattice-based cryptography parameters against known attacks under various security regimes and performance constraints.
- Performance engineering across the proving stack, including sum-check optimization.

Millennium Researcher, June. 2024 - Apr. 2025

- Convex optimization and robust statistical estimation for real-time trading systems.

Stellar Development Foundation Consultant, June. 2023 - Sept. 2023

- Byzantine fault tolerance, C++, stellar-core.
- Improved maximum transaction rate per second by 17% via changes to the consensus protocol, reducing number of communication rounds in overlay network. Formal verification in TLA+.

Distributed Systems and TCS Research, Stanford University Researcher, Sept. 2023 - June. 2024

- Under David Mazières, research in general open membership Byzantine fault tolerance protocols (Github), with applications to certificate authority management and automatic password recovery with hardware security modules.
- Research PCP under Aviad Rubinstein, reconstructed proof of the 2-to-2 Games Conjecture. Proved a result about Convex Continuous Class Complexity.

Topology Research, Stanford University Researcher, July. 2020 - May 2022

- Floer homotopy theory under Mohammed Abouzaid; related the complex cobordism Gysin map to the Grothendieck residue symbol and derived a recursive formula for GromovWitten quantum product corrections.
- Research under Ciprian Manolescu. Floer theory, Seiberg-Witten, knot invariants. Produced new potential counterexamples for the smooth Poincare conjecture.

Selected Papers and Manuscripts

Julius Zhang. *The Relative Trace-Zero Subgroup of the Barreto-Naehrig Curves*, Cryptology ePrint Archive, 2026/1311. [PDF]
Julius Zhang. *Complex Cobordism and Formal Group Law*, Manuscript (Undergraduate Honours Thesis), Stanford University, 2024. [PDF]

Talks

- **p-Ordinary Cohomology for $\mathrm{SL}(2)$ over Number Fields**,
  Hida Theory Seminar, Columbia University, 2026.
- **Construction of Seiberg-Witten-Floer stable homotopy types**,
  Homotopy Theory and Manifolds Seminar, Columbia University, 2026.

Education

Courant Institute of Mathematical Sciences, New York University, computer science *PhD.* starting Sept. 2026

- Verifiable systems and theoretical computer science.

Stanford University, computer science *M.S.* and mathematics *B.A.* Sept. 2019 - June. 2024

- General Wang Yaowu Scholarship.

Skills

Computer Science: Rust, C/C++, Python, formal verification, convex optimization, distributed systems, cryptographic protocol design, performance engineering.

Mathematics: number theory, algebraic geometry, symplectic topology, homotopy theory, Floer theory, gauge theory, concentration inequalities.